PG No. 5 $C_{2\mathrm{h}}$ $2/m$ (b-axis setting) [monoclinic]

Table 1: Representation matrices

Irrep.				
A_g	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}1\end{bmatrix}$
B_g	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}-1\end{bmatrix}$
A_u	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}-1\end{bmatrix}$
B_u	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{010}:\begin{bmatrix}-1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}1\end{bmatrix}$